MSF-CBERT: Gated Multi-Source Fusion of Event Narratives and Quality-Enhanced Structured Attributes for GTD Event Classification

Supplementary Material

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S.0 Overview and protocol

This document accompanies the IEEE BigData 2026 paper and reports the supporting tables referenced there as Supplementary Tables S1–S8, followed by additional tables and figures. All values are computed from the per-run result files distributed in `supplementary/results/`; the scripts that produce every table and figure are in `experiments/`.

Splits. *S*: the ten stratified 70/15/15 splits of the paper (seeds 13, 21, 42, 87, 100, 7, 33, 55, 76, 91). *G*: ten group-disjoint 70/15/15 splits of the same 143 586-event corpus with the same seeds and label stratification. A group is a connected component of events linked by (i) the GTD `related` field, (ii) an identical summary after removing the leading date stamp, normalising whitespace and lower-casing, or (iii) TF-IDF cosine ≥ 0.90 between those texts, with all pairs compared exhaustively. Grouping reads no label and only defines partitions; twenty `StratifiedGroupKFold` folds are assigned 14/3/3 to train/validation/test with the split seed, and no group crosses partitions. All train-fit artefacts (medians, redundancy filters, Cramér's V selection, class weights, scaler, augmentation scope, graph) are re-estimated on each G training partition, and the fail-closed leakage audit passes on every G split. *Chronological*: the single chronological split of the paper.

Systems. Unless stated otherwise all neural systems use ConfliBERT-scr-uncased (256 tokens), the paper's hyperparameters (AdamW 2×10^{-5}, batch 32, at most five epochs, early-stopping patience 2, focal loss $\gamma = 2$, training-only paraphrases), no graph pathway, and validation macro-F1 for checkpoint selection; the test partition is evaluated once per run. Fusion variants share the text pathway $\mathbf{h}_t \in \mathbb{R}^{512}$ and the twelve standardised numerics $\mathbf{x}_t$ (dense vector $\mathbf{d}_t \in \mathbb{R}^{256}$): *gated* (the paper's model, $\mathbf{f} = \boldsymbol{\lambda} \odot \mathbf{h} + (1 - \boldsymbol{\lambda}) \odot \tanh(\mathbf{W}_d \mathbf{d})$); *direct concatenation* $[\mathbf{h}; \mathbf{x}]$; *dense concatenation* $[\mathbf{h}; \mathbf{d}]$; *fixed average* $0.5\,\mathbf{h} + 0.5 \tanh(\mathbf{W}_d \mathbf{d})$; and an *MLP fusion* $[\mathbf{h}; \mathbf{d}] \to 681$ tanh units, whose total of 110 008 227 trainable parameters is within 102 of the gated model (110 008 329). Each is followed by the same linear classifier. Feature-group variants remove the listed inputs from the verbalised categoricals or the dense branch (Table S3). The *restricted* input keeps the narrative, region, country, target type and the vicinity flag; *restricted, first sentence* additionally cuts the narrative to its date stamp and first sentence by a deterministic row-local rule applied identically to all partitions.

Reference runs. The paper's main numbers are the published ten-seed runs. All new neural systems and a same-configuration retraining of MSF-CBERT and text+categoricals were trained on four NVIDIA Quadro RTX 5000 (16 GB) GPUs; fusion and feature-group contrasts are paired against this retraining so that both sides share hardware and software. The published MSF-CBERT and MSF-CBERT+G runs used an NVIDIA A100 and the other published systems Quadro RTX 5000 GPUs; the same-seed difference between the retraining and the published MSF-CBERT runs (-0.51 points, $p = 0.33$; Table S14) is a run-to-run variation estimate. Structured-only baselines and all analyses ran on CPU.

Statistics. Paired differences in test macro-F1 over the ten seeds use the Nadeau–Bengio corrected resampled t-test (two-sided, $n_{\text{test}}/n_{\text{train}} = 0.15/0.70$), which accounts for training-set overlap between random splits. Holm correction is applied within four families: fusion (gated vs. direct concatenation, MLP fusion, fixed average; S), feature dependence (gated vs. without outcome fields, text+categoricals, text+numerics, text only; S), group-disjoint (gated vs. BertGCN-style, text+categoricals, text only; G), and graph (MSF-CBERT+G vs. MSF-CBERT on G; graph without categoricals vs. text only on S). All other comparisons are descriptive and unadjusted. Effect sizes are mean paired differences and Cohen's d_z; the number of seeds favouring each side is always given. A larger mean is not called superior without an adjusted test. S and G splits overlap in their events, so their seeds are neither paired nor independent; S-versus-G changes are reported as differences of ten-seed means without a test.

S.1 Feature availability (Table S1)

GTD records no authorship time for summaries or coded attributes, so availability at an initial report cannot be established from the data. The audit marks each input; the restricted-input setting is a reduced-input sensitivity study of retrospective records, not an estimate of performance at first report.

Table S1: Feature availability and target-leakage audit of all model inputs.

Feature	GTD column	Role	Available at first report	Post-event / label-linked	Treatment in experiments
Narrative	summary	ConfliBERT text input	Not established	Contains post-event content	Kept; first-sentence variant in restricted_first
Date stamp in narrative	summary prefix	Part of text input	Plausible	No	Kept
Region	region_txt	Verbalised category (selected on every seed)	Plausible	No	Kept in restricted
Country	country_txt	Verbalised category	Plausible	No	Kept in restricted
Target type	targtype1_txt	Verbalised category	Plausible but not established	Possibly	Kept in restricted; flagged
Weapon type	weaptype1_txt	Verbalised category	Not established	Strongly label-linked	Dropped in restricted; no_weapon
Perpetrator group	gname	Verbalised category	Often not	Post-event attribution	Dropped in restricted; no_perp
Fatalities	nkill	Dense numeric	No	Outcome	no_outcome; no_casualty; dropped in restricted
Injuries	nwound	Dense numeric	No	Outcome	no_outcome; no_casualty; dropped in restricted
Success	success	Dense numeric	No	Outcome and definition-linked	no_outcome; no_success; dropped in restricted
Extended	extended	Dense numeric	No	Outcome and label-linked	no_outcome; no_extended; dropped in restricted
Property damage	property	Dense numeric	No	Outcome and label-linked	no_outcome; no_property; dropped in restricted
Suicide	suicide	Dense numeric	Not established	Possibly label-linked	Dropped in restricted
Multiple	multiple	Dense numeric	No	Analyst linkage	no_coded; dropped in restricted
Criterion 1	crit1	Dense numeric	No	Analyst inclusion coding	no_coded; dropped in restricted
Criterion 2	crit2	Dense numeric	No	Analyst inclusion coding	no_coded; dropped in restricted
Criterion 3	crit3	Dense numeric	No	Analyst inclusion coding	no_coded; dropped in restricted
Vicinity	vicinity	Dense numeric	Plausible	No	Kept in restricted (only numeric retained)
International	int_any	Dense numeric	No	Analyst coding	no_coded; dropped in restricted
Casualty total	casualties	Removed by training-only redundancy filter on all ten seeds	No	Outcome	Not used
Hostage flag	ishostkid	Removed by training-only redundancy filter on all ten seeds	No	Label-defining	Not used
Weapon sub-type	weapsubtype1_txt	Structured-only baseline: removed by NMI filter; not in neural input	Not established	Label-linked	Not used
Province; city	provstate; city	Structured-only baselines only; never verbalised	Plausible	No	Kept in restricted structured-only baselines

Full justification and uncertainty for each field: feature_availability_audit.csv. GTD gives no authorship time for summaries, so availability at first report cannot be established from the data.

S.2 Gated fusion versus simpler fusion (Table S2)

Table S2: Fusion of the same text and numeric inputs (ten seeds, %). Differences are paired over seeds; Nadeau–Bengio corrected resampled t.

Fusion	Trainable params	Non-encoder params	S macro-F1	Gated − variant (S) [95% CI]	p (Holm)	Seeds gated higher	G macro-F1	Gated − variant (G)
MSF-CBERT (gated fusion)	110,008,329	926,985	85.51 ± 0.87	–	–	–	85.56 ± 0.68	–
Direct concatenation of standardised numerics	109,479,797	398,453	84.89 ± 0.81	+0.61 [-0.27, +1.50]	0.152 (0.456)	9/10	84.00 ± 1.12	+1.56, p=0.018, 9/10
Concatenation of dense numeric vector	109,485,321	403,977	85.66 ± 0.65	-0.16 [-1.10, +0.79]	0.712 (unadj.)	4/10	–	–
MLP fusion, parameter-matched	110,008,227	926,883	85.98 ± 0.96	-0.48 [-1.75, +0.80]	0.419 (0.838)	3/10	85.14 ± 0.76	+0.42, p=0.299, 7/10
Fixed 0.5/0.5 average (gate removed)	109,614,601	533,257	85.83 ± 0.62	-0.32 [-1.41, +0.76]	0.517 (0.838)	4/10	–	–
Text+categoricals	109,479,689	398,345	84.77 ± 0.67	+0.73 [-0.49, +1.96]	0.209 (0.242)	8/10	84.17 ± 0.79	+1.39, p<0.001, 10/10

All variants share the ConfliBERT encoder (109,081,344 parameters), inputs, splits, optimiser, epochs, early stopping and validation-based checkpoint selection. Holm adjustment within the three gating comparisons (concatenation, matched MLP, fixed average); concatenation of the dense vector is descriptive. Text+categoricals (no numeric pathway) is shown for reference; its S contrast belongs to the feature-dependence family and its G contrast to the group-disjoint family.

S.3 Feature-group exclusion and restricted input (Table S3)

Table S3: Feature-group exclusion and restricted-input sensitivity (ten seeds; chronological split: one run).

Setting	Removed inputs	S macro-F1	Variant − full (S) [95% CI]	p	Seeds variant higher	G macro-F1	Variant − full (G)	Chrono macro-F1
Full input (MSF-CBERT, gated)	none (text, 5 verbalised categoricals, 12 numerics)	85.51 ± 0.87	–	–	–	85.56 ± 0.68	–	83.30
Without outcome fields	nkill, nwound, success, extended, property	84.87 ± 0.58	-0.64 [-1.48, +0.21]	0.242 (Holm)	1/10	84.17 ± 0.90	-1.39	83.00
Without success	success	84.90 ± 0.82	-0.61 [-1.71, +0.49]	0.241 (unadj.)	1/10	–	–	–
Without extended	extended	86.16 ± 0.78	+0.65 [-0.39, +1.70]	0.191 (unadj.)	10/10	–	–	–
Without property damage	property	85.88 ± 0.88	+0.37 [-0.54, +1.29]	0.381 (unadj.)	7/10	–	–	–
Without fatalities and injuries	nkill, nwound	85.77 ± 0.83	+0.27 [-0.68, +1.21]	0.536 (unadj.)	5/10	–	–	–
Without analyst-coded fields	crit1-3, multiple, int_any	86.06 ± 0.57	+0.55 [-0.34, +1.44]	0.194 (unadj.)	7/10	–	–	–
Numerics: vicinity only	11 numerics (keeps vicinity)	84.55 ± 0.44	-0.95 [-2.14, +0.23]	0.102 (unadj.)	1/10	–	–	–
Text+categoricals	all 12 numerics	84.77 ± 0.67	-0.73 [-1.96, +0.49]	0.242 (Holm)	2/10	84.17 ± 0.79	-1.39	–
Without perpetrator group	gname	85.66 ± 0.74	+0.16 [-0.73, +1.05]	0.700 (unadj.)	4/10	–	–	–
Without weapon type	weaptype1_txt	80.62 ± 0.48	-4.88 [-5.92, -3.84]	<0.001 (unadj.)	0/10	–	–	–
Text+numerics (no categoricals)	all 5 categoricals	80.40 ± 0.67	-5.11 [-6.16, -4.06]	<0.001 (Holm)	0/10	–	–	–
Text only	all categoricals and numerics	79.63 ± 0.42	-5.87 [-7.06, -4.69]	<0.001 (Holm)	0/10	78.49 ± 0.72	-7.07	76.91
Restricted input	gname, weaptype1_txt, 11 numerics (keeps vicinity)	79.38 ± 0.67	-6.12 [-6.97, -5.28]	<0.001 (unadj.)	0/10	78.45 ± 1.53	-7.11	76.75
Restricted input, first sentence	as restricted; narrative cut to date stamp + first sentence	71.59 ± 0.96	-13.92 [-15.60, -12.24]	<0.001 (unadj.)	0/10	–	–	69.45
Text only, first sentence	all structured inputs; narrative cut to first sentence	71.48 ± 0.86	-14.03 [-15.40, -12.66]	<0.001 (unadj.)	0/10	–	–	–

Reference: same-configuration retraining of MSF-CBERT. Holm adjustment over the four feature-dependence comparisons (outcome fields, numerics, categoricals, all structured inputs); all other rows unadjusted. Restricted input keeps the narrative, region, country, target type and vicinity; it is a reduced-input sensitivity study, not an estimate of performance at first report, because GTD summaries are written after the event. The chronological full-input value is the published MSF-CBERT chronological run.

S.4 Group-disjoint evaluation (Tables S4–S5)

Table S4: Original stratified (S) versus group-disjoint (G) evaluation, same configuration, ten seeds each (%).

System	S macro-F1	G macro-F1	G − S	S acc.	G acc.	S group-balanced F1	G group-balanced F1
MSF-CBERT (gated fusion)	85.51 ± 0.87	85.56 ± 0.68	+0.06	93.88 ± 0.23	93.76 ± 0.24	85.37 ± 0.87	85.64 ± 0.71
Text+categoricals	84.77 ± 0.67	84.17 ± 0.79	-0.60	92.67 ± 0.20	92.20 ± 0.40	84.61 ± 0.73	84.17 ± 0.81
Text only	79.63 ± 0.42	78.49 ± 0.72	-1.14	88.59 ± 0.19	87.98 ± 0.26	79.45 ± 0.47	78.62 ± 0.45
Direct concatenation of standardised numerics	84.89 ± 0.81	84.00 ± 1.12	-0.89	92.81 ± 0.22	92.36 ± 0.47	–	–
MLP fusion, parameter-matched	85.98 ± 0.96	85.14 ± 0.76	-0.84	93.93 ± 0.27	93.73 ± 0.27	–	–
Without outcome fields	84.87 ± 0.58	84.17 ± 0.90	-0.70	92.63 ± 0.20	92.28 ± 0.29	–	–
Restricted input	79.38 ± 0.67	78.45 ± 1.53	-0.93	88.44 ± 0.32	87.75 ± 0.79	–	–

Groups: connected components of GTD related links, identical summaries (date stamp removed, whitespace-normalised, lower-case) and TF-IDF cosine $\geq$0.90 (exhaustive pairwise comparison); 124,616 groups, 115,460 singletons, 28,126 events in multi-event groups, largest 100. G test partitions: 21,537–21,539 events in 18,641–18,705 groups; maximum class-share gap test vs corpus 0.009 points. In S, 16.52–16.96% of test events have a group member in training (0% in G). G − S is a difference of ten-seed means without a test (the splits overlap, so seeds are neither paired nor independent); no events are removed and class shares are held equal by stratification. Group-balanced macro-F1 weights each test event by 1/(size of its group in the test set).

Table S5: Grouping thresholds and exposure: share of test events with a member of their group in the training partition.

Grouping	Groups	Events in multi-event groups	Largest group	S: test exposed % (min–max)	G: test exposed % (min–max)
related + identical summary	126,929	24,889	100	14.51–14.99	0.00–0.00
+ cosine $\geq$0.95	126,283	25,937	100	15.10–15.55	0.00–0.00
+ cosine $\geq$0.90 (primary)	124,616	28,126	100	16.52–16.96	0.00–0.00
+ cosine $\geq$0.85	122,167	31,203	210	18.44–18.97	3.17–3.96
+ cosine $\geq$0.80	118,357	35,642	1076	21.35–22.04	6.66–7.76

Edges accumulate: each row adds near-duplicate pairs at the stated TF-IDF cosine to the related and identical-summary links. G splits are built from the primary (0.90) groups, so their exposure is zero by construction at $\geq$0.90 and non-zero only at looser thresholds. At 0.80, chaining produces a component of 1,076 events, so 0.80 is used for sensitivity only.

S.5 Graph contribution (Table S6)

Table S6: Graph contribution (test macro-F1, points, ten seeds).

Contrast	Mean diff. [95% CI]	p	p Holm	Seeds first higher
MSF-CBERT+G − MSF-CBERT (S, published runs)	-0.17 [-1.15, +0.80]	0.697	–	4/10
Text+graph − text+categoricals (S, published runs)	+0.12 [-0.57, +0.82]	0.696	–	5/10
MSF-CBERT+G − MSF-CBERT (G)	-0.56 [-1.18, +0.07]	0.075	0.075	0/10
Text+graph (no categoricals) − text only (S)	+3.89 [+2.80, +4.98]	<0.001	<0.001	10/10
Text+categoricals − text only (S, reference)	+5.14 [+4.47, +5.80]	<0.001	–	10/10

The graph links perpetrator, target type, weapon type and country, the same fields that are verbalised as categoricals. Published-run rows reproduce the paper's contrasts (unadjusted here).

S.6 Complete baseline comparison (Table S7)

Table S7: Complete baseline comparison over ten seeds (%, mean ± SD). S: original stratified splits; G: group-disjoint splits.

System	S test macro-F1	S test acc.	G test macro-F1	G test acc.
MSF-CBERT (published runs)	86.01 ± 0.67	93.88 ± 0.16	–	–
MSF-CBERT+G (published runs)	85.84 ± 0.89	93.85 ± 0.26	85.01 ± 0.85	93.58 ± 0.23
Text+categoricals (published runs)	84.68 ± 0.81	92.56 ± 0.44	–	–
Text+graph (published runs)	84.81 ± 1.00	92.65 ± 0.16	–	–
BertGCN-style (published runs)	84.99 ± 0.84	93.05 ± 0.37	84.47 ± 0.73	92.68 ± 0.24
MSF-CBERT (gated fusion)	85.51 ± 0.87	93.88 ± 0.23	85.56 ± 0.68	93.76 ± 0.24
Text+categoricals	84.77 ± 0.67	92.67 ± 0.20	84.17 ± 0.79	92.20 ± 0.40
Text only	79.63 ± 0.42	88.59 ± 0.19	78.49 ± 0.72	87.98 ± 0.26
Direct concatenation of standardised numerics	84.89 ± 0.81	92.81 ± 0.22	84.00 ± 1.12	92.36 ± 0.47
Concatenation of dense numeric vector	85.66 ± 0.65	93.79 ± 0.24	–	–
MLP fusion, parameter-matched	85.98 ± 0.96	93.93 ± 0.27	85.14 ± 0.76	93.73 ± 0.27
Fixed 0.5/0.5 average (gate removed)	85.83 ± 0.62	93.88 ± 0.27	–	–
XGBoost, earlier features (ordinal codes)	67.11 ± 1.30	89.17 ± 0.15	–	–
XGBoost, native categoricals, same structured inputs	67.51 ± 1.34	89.10 ± 0.18	65.67 ± 0.58	88.12 ± 0.98
MLP (one-hot), same structured inputs	67.56 ± 1.21	88.66 ± 0.34	66.57 ± 1.33	88.45 ± 0.26

BertGCN-style and MSF-CBERT+G G-columns are the same systems evaluated on the group-disjoint splits. Published runs and same-configuration retraining use identical splits and settings on different GPUs.

S.7 Statistical tests (Table S8)

Table S8: Paired comparisons of test macro-F1 (points): Nadeau–Bengio corrected resampled t (two-sided, $n_{test}/n_{train} = 0.15/0.70$), Holm within each confirmatory family (fusion, feature dependence, group-disjoint, graph).

Family	Comparison	Mean diff. [95% CI]	d_z	p	p Holm	Seeds first higher
Fusion (S)	MSF-CBERT (gated fusion) vs Direct concatenation of standardised numerics	+0.61 [-0.27, +1.50]	0.88	0.152	0.456	9/10
Fusion (S)	MSF-CBERT (gated fusion) vs MLP fusion, parameter-matched	-0.48 [-1.75, +0.80]	-0.47	0.419	0.838	3/10
Fusion (S)	MSF-CBERT (gated fusion) vs Fixed 0.5/0.5 average (gate removed)	-0.32 [-1.41, +0.76]	-0.38	0.517	0.838	4/10
Feature dependence (S)	MSF-CBERT (gated fusion) vs Without outcome fields	+0.64 [-0.21, +1.48]	0.96	0.121	0.242	9/10
Feature dependence (S)	MSF-CBERT (gated fusion) vs Text+categoricals	+0.73 [-0.49, +1.96]	0.76	0.209	0.242	8/10
Feature dependence (S)	MSF-CBERT (gated fusion) vs Text+numerics (no categoricals)	+5.11 [+4.06, +6.16]	6.18	<0.001	<0.001	10/10
Feature dependence (S)	MSF-CBERT (gated fusion) vs Text only	+5.87 [+4.69, +7.06]	6.28	<0.001	<0.001	10/10
Group-disjoint (G)	MSF-CBERT (gated fusion) vs BertGCN-style	+1.10 [+0.35, +1.85]	1.86	0.009	0.009	10/10
Group-disjoint (G)	MSF-CBERT (gated fusion) vs Text+categoricals	+1.39 [+0.74, +2.05]	2.70	<0.001	0.002	10/10
Group-disjoint (G)	MSF-CBERT (gated fusion) vs Text only	+7.07 [+6.29, +7.85]	11.56	<0.001	<0.001	10/10
Graph	MSF-CBERT+G vs MSF-CBERT (gated fusion)	-0.56 [-1.18, +0.07]	-1.13	0.075	0.075	0/10
Graph	Text+graph (no categoricals) vs Text only	+3.89 [+2.80, +4.98]	4.53	<0.001	<0.001	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs Concatenation of dense numeric vector	-0.16 [-1.10, +0.79]	-0.21	0.712	–	4/10
descriptive (S)	MSF-CBERT (gated fusion) vs Numerics: vicinity only	+0.95 [-0.23, +2.14]	1.02	0.102	–	9/10
descriptive (S)	MSF-CBERT (gated fusion) vs Restricted input	+6.12 [+5.28, +6.97]	9.20	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs Restricted input, first sentence	+13.92 [+12.24, +15.60]	10.50	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs Text only, first sentence	+14.03 [+12.66, +15.40]	12.98	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without weapon type	+4.88 [+3.84, +5.92]	5.93	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without perpetrator group	-0.16 [-1.05, +0.73]	-0.22	0.700	–	6/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without analyst-coded fields	-0.55 [-1.44, +0.34]	-0.79	0.194	–	3/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without extended	-0.65 [-1.70, +0.39]	-0.79	0.191	–	0/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without property damage	-0.37 [-1.29, +0.54]	-0.52	0.381	–	3/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without fatalities and injuries	-0.27 [-1.21, +0.68]	-0.36	0.536	–	5/10
descriptive (S)	MSF-CBERT (gated fusion) vs Without success	+0.61 [-0.49, +1.71]	0.70	0.241	–	9/10
descriptive (S)	MSF-CBERT (gated fusion) vs XGBoost, earlier features (ordinal codes)	+18.39 [+16.84, +19.95]	15.01	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs XGBoost, native categoricals, same structured inputs	+17.99 [+16.11, +19.88]	12.10	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs MLP (one-hot), same structured inputs	+17.95 [+16.20, +19.70]	13.03	<0.001	–	10/10
descriptive (S)	Restricted input vs XGBoost, native categoricals, restricted input	+54.87 [+54.14, +55.60]	95.32	<0.001	–	10/10
descriptive (S)	Restricted input vs MLP (one-hot), restricted input	+54.09 [+52.18, +55.99]	35.91	<0.001	–	10/10
descriptive (S)	Without outcome fields vs XGBoost, native categoricals, without outcome fields	+27.46 [+25.63, +29.30]	19.01	<0.001	–	10/10
descriptive (S)	Restricted input vs Restricted input, first sentence	+7.79 [+6.61, +8.98]	8.32	<0.001	–	10/10
descriptive (S)	MSF-CBERT (gated fusion) vs MSF-CBERT (published runs)	-0.51 [-1.63, +0.61]	-0.57	0.332	–	3/10
descriptive (S)	Text+categoricals vs Text+categoricals (published runs)	+0.09 [-0.59, +0.77]	0.16	0.776	–	5/10
descriptive (S)	MSF-CBERT (published runs) vs BertGCN-style (published runs)	+1.02 [-0.04, +2.09]	1.22	0.058	–	8/10
descriptive (S)	MSF-CBERT (published runs) vs Text+graph (published runs)	+1.20 [+0.14, +2.27]	1.43	0.031	–	9/10
descriptive (S)	MSF-CBERT (published runs) vs Text+categoricals (published runs)	+1.33 [+0.25, +2.41]	1.56	0.021	–	10/10
descriptive (S)	MSF-CBERT+G (published runs) vs MSF-CBERT (published runs)	-0.17 [-1.15, +0.80]	-0.23	0.697	–	4/10
descriptive (G)	MSF-CBERT (gated fusion) vs Direct concatenation of standardised numerics	+1.56 [+0.34, +2.78]	1.63	0.018	–	9/10
descriptive (G)	MSF-CBERT (gated fusion) vs MLP fusion, parameter-matched	+0.42 [-0.45, +1.29]	0.62	0.299	–	7/10
descriptive (G)	MSF-CBERT (gated fusion) vs Without outcome fields	+1.39 [+0.64, +2.15]	2.33	0.002	–	10/10
descriptive (G)	MSF-CBERT (gated fusion) vs Restricted input	+7.11 [+5.64, +8.59]	6.12	<0.001	–	10/10
descriptive (G)	MSF-CBERT (gated fusion) vs XGBoost, native categoricals, same structured inputs	+19.90 [+18.91, +20.89]	25.50	<0.001	–	10/10
descriptive (G)	MSF-CBERT (gated fusion) vs MLP (one-hot), same structured inputs	+19.00 [+16.84, +21.16]	11.14	<0.001	–	10/10
descriptive (G)	Restricted input vs XGBoost, native categoricals, restricted input	+54.30 [+52.04, +56.57]	30.41	<0.001	–	10/10
descriptive (G)	Restricted input vs MLP (one-hot), restricted input	+54.47 [+51.21, +57.73]	21.16	<0.001	–	10/10

Effect size d_z = mean paired difference / SD of differences. Descriptive rows are unadjusted. The corrected test accounts for training-set overlap between random splits but remains approximate with ten seeds.

S.8 Additional tables

Table S9: Per-class test accuracy of MSF-CBERT trained on the original splits, for test events with and without a group member in training (pooled over ten seeds, %).

Attack type	Unexposed n (pooled)	Exposed n (pooled)	Acc. unexposed	Acc. exposed	Exposed − unexposed
Armed Assault	44,498	6,582	89.49	91.90	+2.41
Assassination	13,578	492	87.97	89.02	+1.05
Bombing/Explosion	85,256	20,599	96.71	97.73	+1.02
Facility/Infrastructure Attack	9,056	3,347	90.28	95.76	+5.47
Hijacking	563	84	73.89	71.43	-2.46
Hostage Taking (Barricade Incident)	786	88	57.89	73.86	+15.98
Hostage Taking (Kidnapping)	14,784	1,846	94.04	95.34	+1.30
Unarmed Assault	1,115	201	85.83	96.02	+10.19
Unknown	9,648	2,857	96.99	98.39	+1.40

Same-configuration retraining. Pooled overall accuracy: MSF-CBERT (gated fusion) 93.42 unexposed vs 96.17 exposed; published MSF-CBERT runs 93.43 unexposed vs 96.15 exposed; Text+categoricals 92.03 unexposed vs 95.86 exposed; Text only 87.89 unexposed vs 92.04 exposed. Descriptive; exposed and unexposed events differ in class mix and difficulty, so no subset macro-F1 is claimed.

Table S10: Per-class test F1 on original (S) and group-disjoint (G) splits, same configuration, ten seeds (%, mean ± SD).

Attack type	S MSF-CBERT	S Text+cat.	G MSF-CBERT	G Text+cat.
Armed Assault	90.98 ± 0.21	90.35 ± 0.24	90.74 ± 0.31	89.79 ± 0.68
Assassination	81.05 ± 1.24	74.28 ± 0.66	81.21 ± 1.07	73.37 ± 1.26
Bombing/Explosion	97.94 ± 0.16	97.06 ± 0.12	97.91 ± 0.17	96.79 ± 0.23
Facility/Infrastructure Attack	88.97 ± 0.58	88.59 ± 0.86	88.07 ± 0.41	87.45 ± 0.99
Hijacking	73.15 ± 3.65	73.01 ± 4.04	74.14 ± 3.87	72.82 ± 3.23
Hostage Taking (Barricade Incident)	61.95 ± 3.65	63.51 ± 3.92	63.53 ± 4.23	64.23 ± 4.16
Hostage Taking (Kidnapping)	93.69 ± 0.45	93.87 ± 0.28	93.66 ± 0.52	93.58 ± 0.43
Unarmed Assault	85.30 ± 2.35	86.11 ± 2.00	84.36 ± 2.04	83.48 ± 2.31
Unknown	96.51 ± 0.38	96.17 ± 0.51	96.46 ± 0.32	96.04 ± 0.37

Table S11: Chronological split (train to 2016-08-18, test from 2018-07-09), one run per system (%).

System	Val macro-F1	Test macro-F1	Test acc.
Text only	80.09	76.91	84.30
Without outcome fields	86.29	83.00	92.67
Restricted input	80.03	76.75	85.03
Restricted input, first sentence	70.75	69.45	79.81
MSF-CBERT (published run)	87.63	83.30	92.94
Text+categoricals (published run)	85.59	83.23	92.59

Single split: descriptive only, no test.

Table S12: XGBoost on restricted structured inputs: source of the large accuracy SD across seeds.

Split	Validation-selected class weighting	Seeds	Val macro-F1	Test macro-F1	Test acc.	Predicted Bombing share %
G	none	3	24.50	23.42	55.24	69.4
G	balanced	7	24.19	24.46	35.76	22.4
S	none	4	25.00	24.51	55.32	69.0
S	balanced	6	24.53	24.50	35.89	22.6

Validation macro-F1 is nearly tied between class-balanced and unweighted training (differences below one point), so the validation-selected setting alternates across seeds. Balanced weighting predicts the majority class far less often and lowers accuracy by about 19 points at equal macro-F1. Reported as selected on validation; not re-selected on test.

Table S13: Group-disjoint partitions: sizes, groups per partition and largest test-vs-corpus class-share difference.

Seed	Train events	Val events	Test events	Train groups	Val groups	Test groups	Max class-share gap (pts)
13	100,512	21,537	21,537	87,212	18,702	18,702	0.006
21	100,510	21,537	21,539	87,275	18,640	18,701	0.004
42	100,508	21,539	21,539	87,220	18,695	18,701	0.004
87	100,512	21,537	21,537	87,209	18,706	18,701	0.005
100	100,510	21,538	21,538	87,216	18,695	18,705	0.003
7	100,511	21,537	21,538	87,211	18,702	18,703	0.006
33	100,511	21,537	21,538	87,273	18,702	18,641	0.003
55	100,510	21,537	21,539	87,212	18,701	18,703	0.009
76	100,510	21,537	21,539	87,210	18,702	18,704	0.007
91	100,512	21,537	21,537	87,276	18,642	18,698	0.007

Twenty StratifiedGroupKFold folds (label-stratified, grouped by the primary 0.90 groups) assigned 14/3/3 to train/validation/test with the split seed. No group crosses partitions (asserted). Class counts per partition: groupseed_split_report_*.json.

Table S14: Manuscript claims versus verified and new results.

Claim	Manuscript	Verified / new	Source
MSF-CBERT test macro-F1 (S)	86.01 $\pm$ 0.67	86.01 $\pm$ 0.67 recomputed; retraining 85.51 $\pm$ 0.87	per_run_metrics.csv
MSF-CBERT accuracy (S)	93.88 $\pm$ 0.16	93.88 $\pm$ 0.16	per_run_metrics.csv
MSF-CBERT+G (S)	85.84 $\pm$ 0.89	85.84 $\pm$ 0.89	per_run_metrics.csv
Text+categoricals (S)	84.68 $\pm$ 0.81	84.68 $\pm$ 0.81; retraining 84.77 $\pm$ 0.67	per_run_metrics.csv
Text+graph (S)	84.81 $\pm$ 1.00	84.81 $\pm$ 1.00	per_run_metrics.csv
BertGCN-style (S)	84.99 $\pm$ 0.84	84.99 $\pm$ 0.84	per_run_metrics.csv
Chronological MSF-CBERT	83.30	83.30	per_run_metrics.csv
Retraining − published MSF-CBERT (run-to-run)	–	-0.51, p=0.332, 3/10	contrasts.csv
Test events sharing a group with training (S)	3.05% verbatim, 3.30% near-dup.	16.52–16.96% (related, identical, cosine $\geq$0.90)	group_exposure_summary.csv
MSF-CBERT on group-disjoint splits	–	85.56 $\pm$ 0.68	arm_summary.csv
Gate vs simpler fusion	gated branch (no comparison)	no measurable difference (Holm p $\geq$ 0.46)	contrasts.csv (fusion family)
Graph redundancy	not isolated	graph without categoricals +3.89 over text only; +G on G −0.56 (p=0.075)	contrasts.csv (graph family)
Unknown categorical values	explicit Unknown states	kept as Unknown in data; verbalisation omits the phrase	src/features/verbalize.py

S.9 Figures

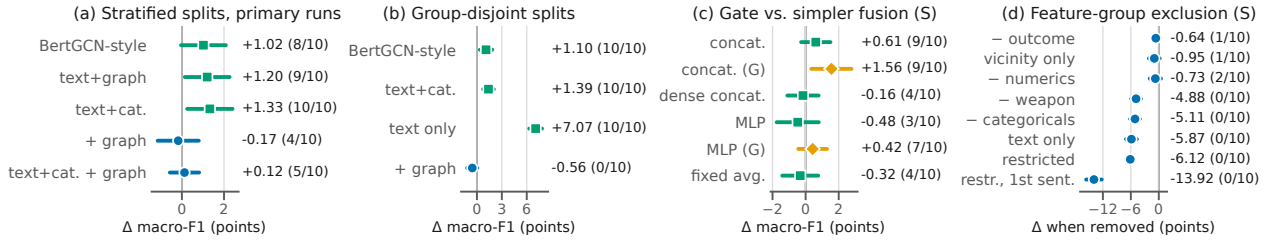


Figure S1: Paired ten-seed differences in test macro-F1 (mean, 95% CI, seeds on which the first system scored higher): (a) primary stratified-split runs of the paper, (b) group-disjoint splits, (c) gated versus simpler fusion of the same inputs, (d) change when input groups are removed. Green squares: MSF-CBERT vs. comparison system; blue circles: component added or removed; amber diamonds in (c): same contrast on group-disjoint (G) splits. This is Fig. 6 of the paper.

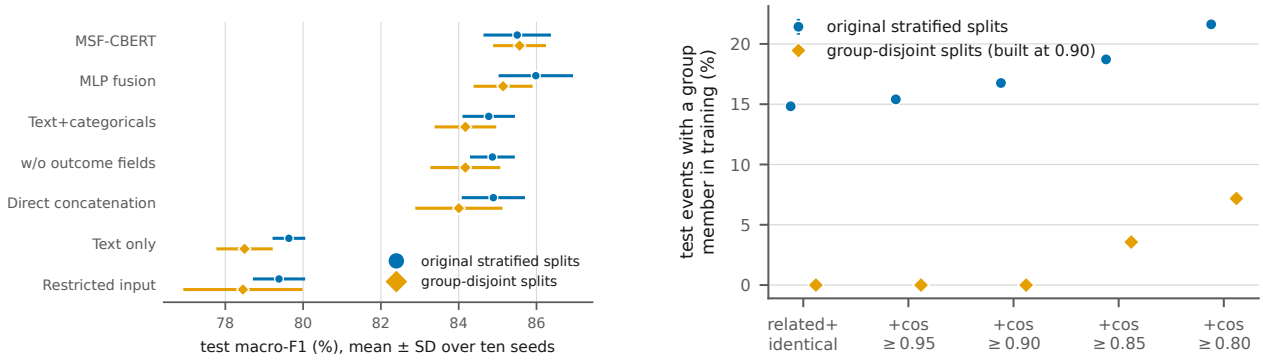


Figure S2: Left: test macro-F1 on original stratified and group-disjoint splits for the same configurations (mean ± SD, ten seeds; descriptive). Right: share of test events with a member of their group in training under each grouping threshold (mean, range over ten seeds).

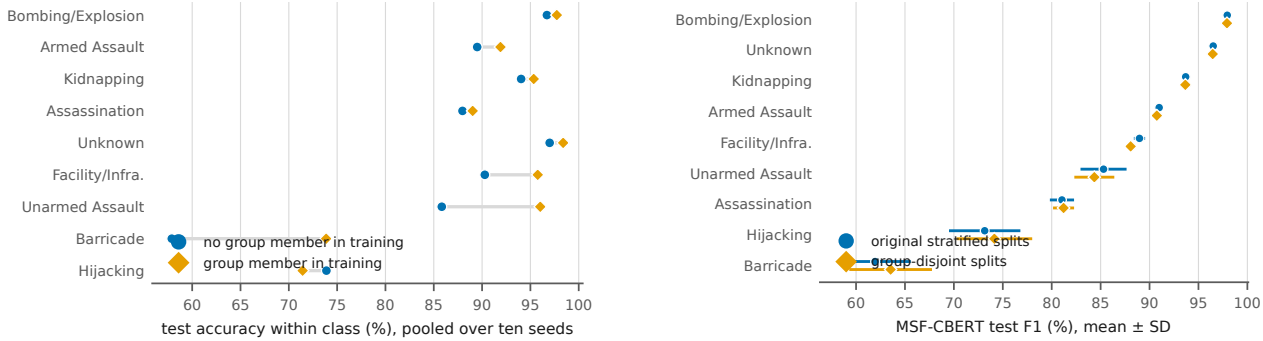


Figure S3: Left: per-class accuracy of MSF-CBERT trained on the original splits for test events with and without a group member in training (pooled over ten seeds; descriptive, classes differ in difficulty and mix). Right: per-class test F1 of MSF-CBERT on original and group-disjoint splits (mean ± SD).

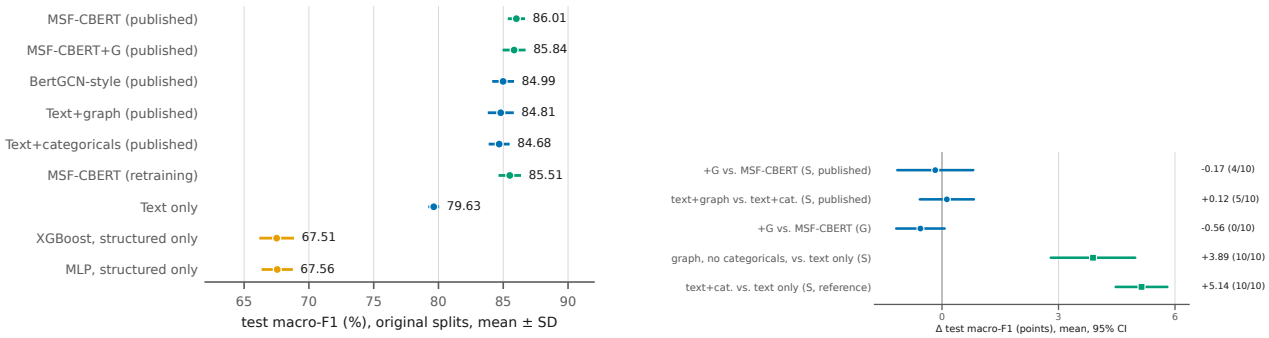


Figure S4: Left: test macro-F1 of all systems on the original splits including structured-only baselines (mean ± SD). Right: graph contrasts (paired differences, mean, 95% CI, seeds on which the first system scored higher).